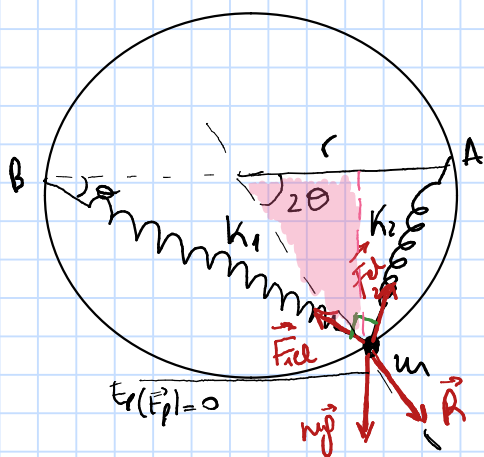


energia potenziale $\rightarrow$ funzione di stato



k_1, k_2, m, r
quali sono i punti di equilibrio?

teorema della corda

$$E_q = \sum \vec{F} = 0$$

$$\vec{F}_{el} + \vec{F}_{el} + \vec{R} + m\vec{g} = 0$$

Con energia:

$$E_M = E_c + E_p = \text{costante}$$

solo forze conservative

Poi E_p trovo i minimi e

lo faccio imponendo la sua derivata a zero

$$\overline{AC} = 2r \sin(\theta)$$

$$\overline{BC} = 2r \cos(\theta)$$

$$E_p = E_{p1} + E_{p2} + E_{p(g)} = \frac{1}{2} k_1 \overline{AC}^2 + \frac{1}{2} k_2 \overline{BC}^2 + m g r (1 - \sin(2\theta))$$

$$= \frac{1}{2} k_1 4r^2 \sin^2(\theta) + \frac{1}{2} k_2 4r^2 \cos^2(\theta) + m g r (1 - \sin(2\theta))$$

$$= 2r^2 \left(k_1 \frac{1 - \cos(2\theta)}{2} + k_2 \frac{1 + \cos(2\theta)}{2} \right) + m g r - m g r \sin(2\theta)$$

$$= r^2 (k_1 + k_2 + \cos(2\theta)(k_2 - k_1)) + m g r - m g r \sin(2\theta)$$

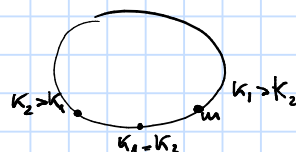
$$= \underbrace{(m g r + (k_1 + k_2) r^2)}_0 + \underbrace{(k_2 - k_1) r^2 \cos(2\theta) - m g r \sin(2\theta)}_{A \sin(\varphi - 2\theta)} =$$

c'è un modo per scrivere quella cosa in quel modo per semplificare il dover fare una derivata vedo che posso fare perché ho la combinazione di seno e coseno con stesso argomento

$$E_p - \text{costante} = \underbrace{(k_2 - k_1) r^2 \cos(2\theta)}_a - \underbrace{m g r \sin(2\theta)}_b = a \cos(2\theta) - b \sin(2\theta) = b \left(\frac{a}{b} \cos(2\theta) - \sin(2\theta) \right) =$$

$$= m g r \underbrace{\left(\frac{k_2 - k_1}{m g} r \right)}_A \cos(2\theta) - \sin(2\theta)$$

se $k_2 > k_1 \rightarrow$ positiva
se $k_2 < k_1 \rightarrow$ negativa
se $k_2 = k_1 \rightarrow$ p.to di equilibrio



$$= m g r \sqrt{1 + A^2} \left(\frac{A}{\sqrt{1 + A^2}} \cos(2\theta) - \frac{1}{\sqrt{1 + A^2}} \sin(2\theta) \right)$$

$$\frac{A^2}{(\sqrt{1 + A^2})^2} + \frac{1^2}{(\sqrt{1 + A^2})^2} = 1 \quad \text{si comportano come il seno e il coseno}$$

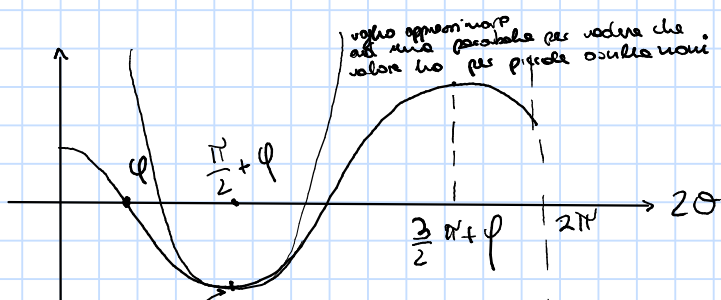
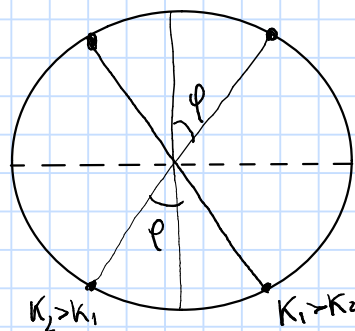
$$\sin^2 x = \frac{\tan^2(x)}{1 + \tan^2(x)}$$

$$\cos^2 x = \frac{1}{1 + \tan^2(x)}$$

$$E_p(\theta) = m g r \sqrt{1 + A^2} (\sin(\varphi) \cos(2\theta) - \cos(\varphi) \sin(2\theta)) =$$

$$= m g r \sqrt{1 + A^2} \sin(\varphi - 2\theta) = m g r \sqrt{1 + A^2} \sin(2\theta - \varphi)$$

$$\varphi = \arctan\left(\frac{(k_2 - k_1)r}{m g}\right)$$



un minimo

$$E_p = E_p(\theta_{eq}) + E_p'(\theta_{eq})(\theta - \theta_{eq}) + \frac{1}{2} E_p''(\theta_{eq})(\theta - \theta_{eq})^2 + \dots$$

